Scalability & Future Plan

Date	13 July 2026
Team ID	XXXXXX
Project Name	Personalized Networking Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No user login/authentication system implemented	Any user can access the application without identity verification; no personalized user accounts or data privacy	High
2	No proper database (e.g., MySQL) implemented; data stored in local JSON files (history.json, feedback.json)	Data is not structured, difficult to query, prone to file corruption, and not scalable for multiple users	High
3	High response time (~44 seconds average) due to zero-shot classification model inference	Slows down user experience; not suitable for real-time interaction	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles requests sequentially; tested with 10 concurrent iterations	Implement asynchronous request handling and load balancing for multiple simultaneous users
2	Data Storage	Uses local JSON files (history.json, feedback.json) for data storage	Migrate to a database (e.g., PostgreSQL/MongoDB) for better scalability and data integrity
3	Performance	Average response time ~44 seconds due to model inference on CPU	Deploy model on GPU-enabled server or use a lighter/distilled model for faster inference
4	Security	Basic local API with no authentication implemented	Add API key authentication and rate limiting to prevent misuse in production

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Migrate data storage from JSON files to a proper database	1 month	Improved data reliability and scalability
Phase 3	Optimize model inference (GPU deployment / model caching)	2 months	Reduce response time from ~44 sec to under 5 sec
Phase 4	Add user authentication and multi-user support	3 months	Enable secure, personalized experience for multiple users